

TALOS

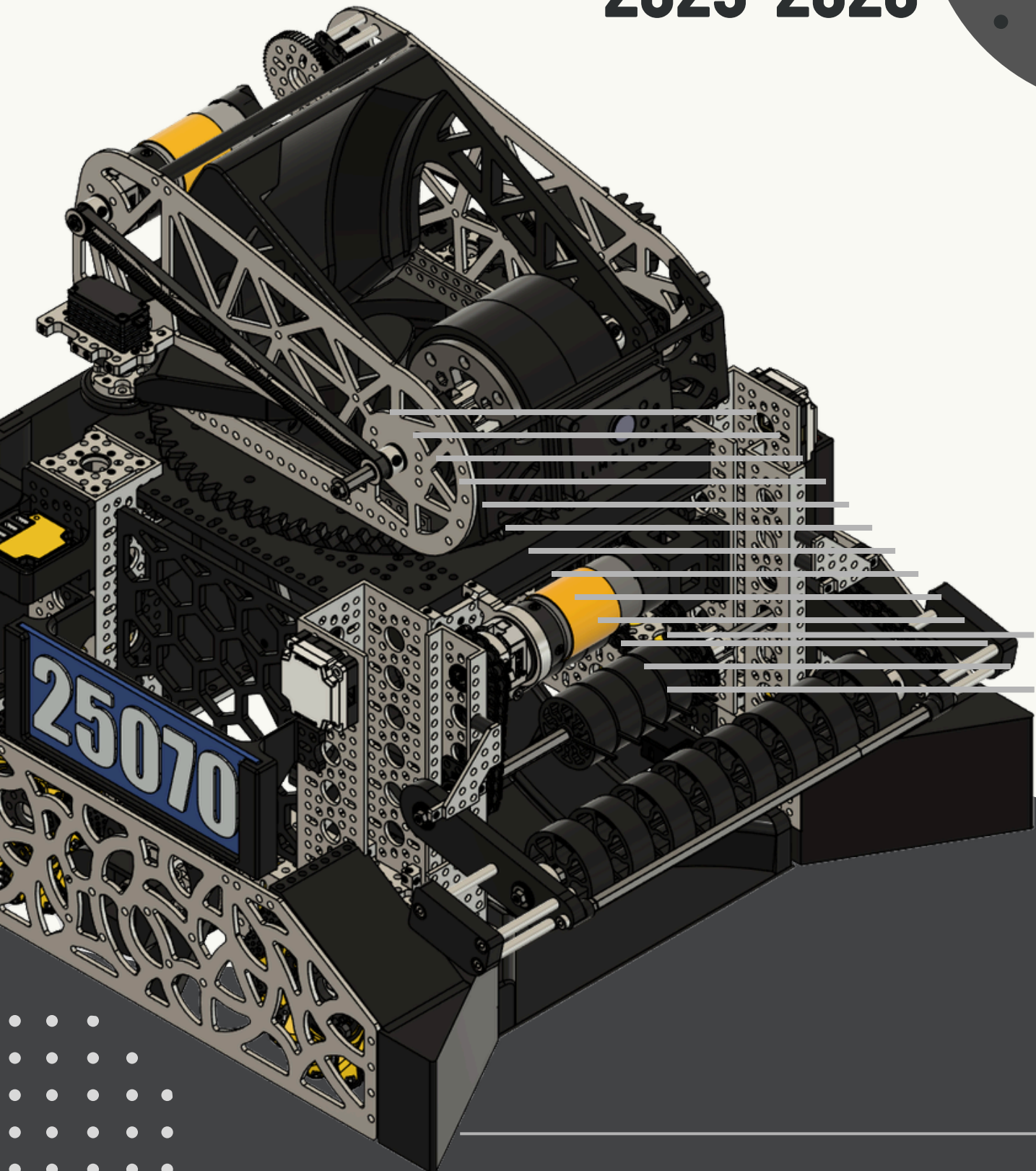
#25070

EDUROBOTS



Engineering
Portfolio

2025-2026



INTRODUCTION

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We are **FTC Team Talos EduRobots 25070**, a student robotics team from **Ierapetra, Crete**, founded in **2023** during the **Centerstage season** and hosted by **EduRobots**. **Talos**, the mythological guardian of **Crete**, inspired our name.

Season Goal: Improve after Nationals and prepare for Istanbul Premier Event 2026

MEET OUR MEMBERS

ENGINEERS

Leader



George



Eirini



Manos L.

DESIGNERS

Leader



Stathis



Thanasis

CODERS

Leader



Manos D.



Ioanna

OUTREACH

Leader



Kyveli



Nikos

Our team is organized into **specialized departments**, allowing each member to contribute through their strengths while keeping **communication** and **collaboration** strong across the whole team.

EVOLUTION

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TURNING POINT



Season Kickoff

FTC Greece Nationals

Instabul Premier Event

CHALLENGES

- Department Coordination → Inconsistent under competitive pressure
 - Documentation → Lacked in depth technical analysis
 - Presentation → Vague explanations and no actual team evaluation opportunities
-
- Engineering-Strategic choices → Not intertwined and a lack of connection between them.
 - Strategy Philosophy → No realistic, targeted goals
 - Programming

Non-Technical

Technical

IMPROVEMENTS

- Department Coordination → Weekly inter-department meetings for coordinating
 - Documentation → Rethinking portfolio-banner philosophy
 - Presentation → Including data, in-depth decision-making criteria, and pointing out our strengths
-
- Engineering-Strategic choices → Aligning our design priorities with our team's goal.
 - Strategic Philosophy → Decided to craft a new strategy with realistic goals
 - Programming

Building a working robot → Creating a reliable engineering process.

STRATEGY

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Our main strategic priority this season was to consistently secure the **Goal Ranking Point** by **maximizing artifact scores**. After detailed **game analysis**, we determined that rapid, reliable shooting offered the optimal balance of **points, consistency, and alliance contribution**. As a result, our robot was engineered for **swift artifact intake, stable transfer, precise shooting, and autonomous adaptability**.

Target: 36+ artifacts per match as an alliance

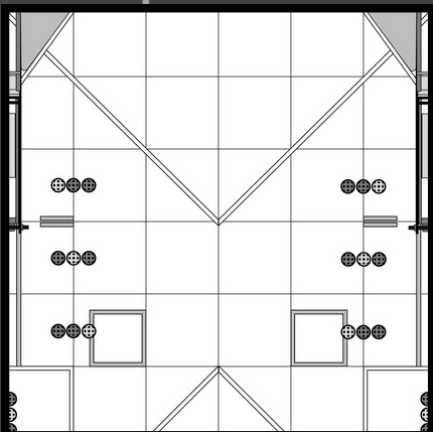
Since our robot can usually score around **12 artifacts in autonomous**, our **tele-op strategy** focuses on completing the remaining scoring requirement as efficiently as possible with our **alliance partner**.

Autonomous

We created **four autonomous routines: close and far starts for both alliances**. We favor the launch zone **near the goal** for faster, more consistent scoring, but keep a far-side option for **alliance partner strategies**.

Specifically, we:

- Start with preloaded artifacts.
- Collect field artifacts as needed.
- Shoot the gathered artifacts.
- Adjust path for partner's specific requests.



TeleOp

During **tele-op**, we focus on **rapid artifact cycling, collecting and shooting quickly with minimal alignment errors**. We choose **speed over complex patterns** when **Goal RP** is a better strategy.

Specifically, we:

- Use fast intake cycles.
- Aim assisted by the Turret.
- Use a driver-friendly artifact transfer.
- Prioritize scoring volume.

Endgame

In **endgame**, we attempt to contribute to **Movement RP** opportunities when they do not reduce our main scoring objective. Our **lifting mechanism** also contributes to ranking point gain.

Specifically, we:

- Move toward base in final seconds.
- Use lifting mechanism when useful.
- Support alliance partner positioning.
- Do not sacrifice Goal RP unnecessarily.

DESIGN PRIORITIES

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How Strategy Became Robot Requirements:

Strategic Need	Requirement	Robot Feature
Score many artifacts	Fast cycle time	Active intake & Rapid transfer
Reduce missed shots	Accurate aiming	Turret & hood & velocity control
Score in auto	Repeatable autonomous	Mirrored close & far routines
Adapt to partners	Flexible paths	Multiple autonomous programs
Reduce failures	Strong mechanisms	Redesigned transfer system, & gear control
Save match time	Easy driving	Intake compliance & automated targeting

Strategic Tradeoffs:

We chose **speed and repeatability** over unnecessary complexity. Although **pattern and movement ranking points** are valuable, our most reliable contribution was **high artifact scoring**. This decision shaped every major subsystem: the intake had to be forgiving, the shooter had to recover quickly, the turret had to aim consistently, and autonomous had to work from multiple starting positions.

**Pattern
Perfection** 

**Speed &
Quantity** 

**Complex
Mechanisms** 

**Reliable &
Consistent** 

**Fixed Auto
Routine** 

**Alliance
Flexibility** 

The diagram illustrates the mechanical components of the robot. On the left, the **Transfer System** is shown, featuring a large curved arm with a **Flicker** mechanism at its base. On the right, the **Shooter** is depicted, which includes a **Hood** and a **Turret**. A red LEGO Technic plate with the number **25070** is visible on the base of the shooter. Below the diagrams, a sequence of five rounded rectangular boxes represents the robot's workflow: **Field**, **Suspension Intake**, **Transfer System**, **Shooter**, and **Goal**. Arrows indicate the flow from Field to Suspension Intake, and from Suspension Intake to Transfer System, Shooter, and Goal.

SUBSYSTEM	ROLE IN ARCHITECTURE
Drivetrain	Moves quickly while keeping the robot stable
Suspension Intake	Collects artifacts even with imperfect alignment
Active Transfer System	Moves artifacts predictably toward the shooter
Shooter	Controls speed, compression, and trajectory
Turret	Aims without requiring full robot repositioning
Lift	Adds match flexibility in final seconds

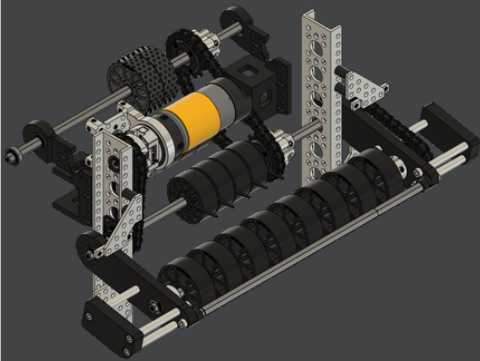
SUBSYSTEM INTEGRATION⁶

HOW COLLECTION, TRANSFER, AND SHOOTING WORK AS ONE SYSTEM

Collection

Active Suspension Intake

The intake uses compliant, 3D-printed **Gecko wheels** and **gravity-assisted tension** to collect artifacts at imperfect angles. This makes the first stage of the cycle more consistent and **forgiving for the driver**.



Faster collection means **shorter cycle time** and **fewer missed pickup attempts**.

Transfer

Active Transfer System

After collection, the active transfer moves the artifact toward the shooter in a **predictable path** using **intake-parallel rows** powered by the **intake motor**.

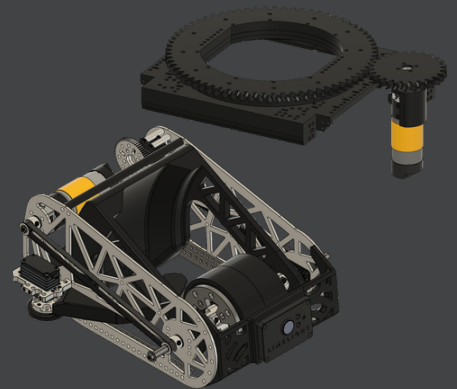


Controlled transfer **reduces jams** and gives the shooter a more **consistent feed**.

Scoring

Turreted Shooter System

The scoring system combines the flywheel shooter, adjustable hood, and gear-driven turret. The flywheel provides launch speed, the hood controls compression and trajectory, and the turret aims without requiring the drivetrain to fully rotate.



The robot can score faster because aiming and shooting are supported mechanically.

WHY THIS ARCHITECTURE WORKS:

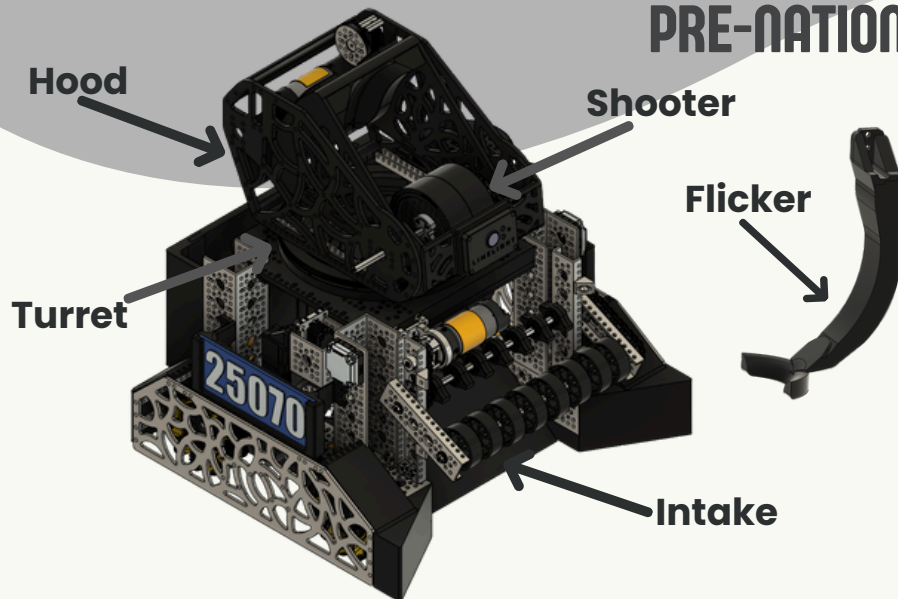
Our design **reduces uncertainty** throughout the cycle. The **intake tolerates imperfect angles**, the **transfer system directs artifacts**, and the **turret ensures aiming consistency**. These combined systems enable fast, reliable scoring.

STAGE	RISK	DESIGN RESPONSE
Collection	Driver misses artifact	Compliant active intake
Transfer	Artifact jams or misfeeds	Powered active transfer
Shooting	Robot not perfectly aligned	Turret + adjustable hood

DESIGN EVOLUTION

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PRE-NATIONALS ROBOT ANALYSIS



Nationals were a **turning point**. Before, our robot could score, but competition exposed issues with **reliability, transfer and shooter consistency**. Afterward, we rebuilt it for better control, replacing major subsystems instead of making small tweaks.

SYSTEM	BENEFITS	OBSTACLES	CONSEQUENCE
Intake	Could collect artifacts	Small range of grip	Inconsistent collection
Flicker Mechanism	Moved artifacts toward shooter	Misfeeds or inconsistent artifact position	Missed shots or delays
Shooter	Could score when fed correctly	Compression and feeding inconsistency affected aim	Lower repeatability
Turret	Allowed aiming	Gear skipping and not alligning	Slower aiming, mechanical stress
Chassis	Supported robot structure	Stability issues appeared under match pressure	Harder repairs

The **pre-Nationals** robot proved our strategy was strong, but showed that the architecture needed improvement. The artifact was not consistently controlled throughout the full scoring path, so after Nationals, we redesigned the robot around a more **predictable intake-transfer-shooter pipeline**. It became the prototype that revealed our next engineering requirements.

The biggest issues: Speed+Consistency

POST-NATIONALS REDESIGN⁸

Instead of only repairing failures, we used Nationals as a redesign checkpoint. We changed the robot's architecture so the artifact would be controlled at every stage: collection, transfer, alignment, and shooting.

AREA	PRE-NATIONALS	POST-NATIONALS	REASON FOR CHANGE
Robot architecture	Mechanisms worked more independently	Connected scoring pipeline	Improve cycle consistency
Intake	Less forgiving pickup	Wider range of grip	Reduce driver alignment errors
Transfer	Flicker worked inconsistency	Active transfer system	Reduce jams and misfeeds
Shooter feed	Artifact position varied	More controlled feed into shooter	Improve shot repeatability
Compression	Compression needed improvement	Modified hood geometry	More consistent trajectory
Turret motion	Gear skipping	Smoother controlled motion	Protect gears and improve aiming
Structure	Match stress revealed weak points	Stronger mounts and protection	Increase reliability

The biggest improvement after Nationals was changing from a robot that could score to a robot that could **score repeatedly under competition pressure**.

Capable→Competitive

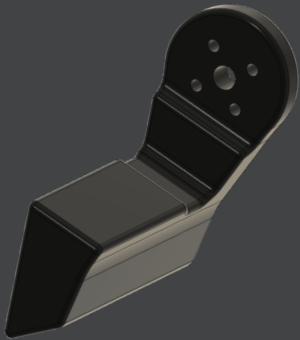
After Nationals, the redesign changed the robot's entire **scoring logic**. The new architecture connected **intake, active transfer, and shooting** into one **controlled pipeline**, with the active transfer system acting as the **key link that made artifact movement more predictable**.

SUBSYSTEM ITERATIONS

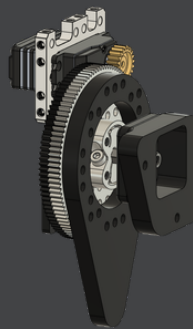
Each subsystem underwent changes following testing and competition feedback. We used the same process for every redesign: **identify the problem, change the design, and evaluate** whether the robot became more **reliable, faster, or easier to control**.

Lift

The lift was less sturdy and had a higher risk of breaking during operation.

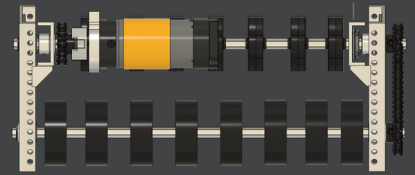


We improved its stability by adding one support on each side and refining the shape.

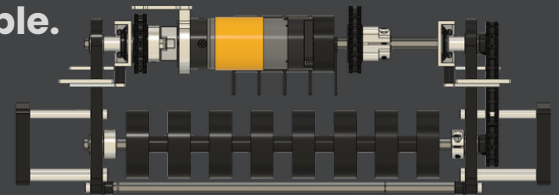


Intake

Our intake had a limited grip range, making it harder to collect artifacts consistently.



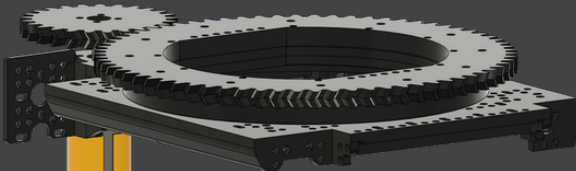
We later extended it to improve grip and make corner pickups more reliable.



Turret

The turret used a small standard toothed gear, which made it more likely to skip under load.

We switched to a larger herringbone gear to improve engagement and prevent skipping.



Transfer

The flickers were slow and inconsistent, which sometimes caused feeding errors.



We improved it into a fast, direct transfer that moves artifacts smoothly into the shooter.



Test-→Problem-→Redesign-→Retest-→Finalize

PROGRAMMING

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CONTROL SYSTEMS

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DATA & TESTING

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Our philosophy is built on :

active participation

shared learning

inspiring people

underepresented groups:

women in STEM

Meeting WOMEN DO BUSINESS organisation

Had the opportunity to meet with Women Do Business, exploring leadership, equality, and women's empowerment in STEM and robotics



FTC teams with limited funding

Cretan scrimmage event 2026

Hosted the first Cretan Scrimmage, giving FTC teams a chance to practice, improve, and collaborate before the competition finals.



international community

FTCxLive seminar 2026

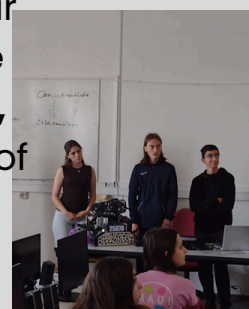
Launched the **FTCxLive seminar** series, to promote collaboration, learning, and connection within the international FTC community. Open participation



Students with Limited STEM Opportunities

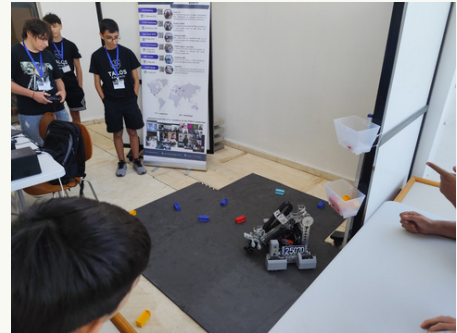
Small-town school visit

Present our FTC journey, showcase our robot, and **introduce students to robotics, STEM**, and the values of FIRST through an interactive presentation.



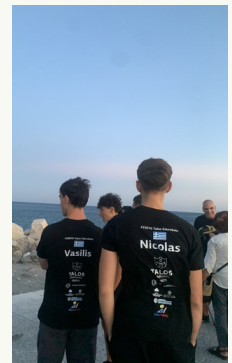
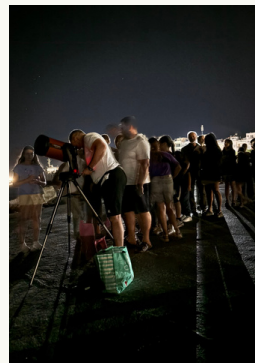
**Visit at
kindergarten**

**Researchers' Night
2025
4k+ audience(all
ages)**



**Total Lunar Eclipse
OBSERVATORY 2025**

**1k+ audience(all
ages)**



SUSTAINABILITY

FUNDING

25.282€

Eugenides Foundation	3.000 €
Holy Metropolis of Ierapytna and Siteia	3.000 €
Hellenic Parliament	2.000€
Ierapetra Municipality	1.750 €
Plastika Kritis	1.500€
Angelicoussis Foundation	1.500€
Eurobank	1.000€
Gene Haas Foundation	500€
Other Companies	11.032 €



ΙΔΡΥΜΑ ΕΥΓΕΝΙΔΟΥ



HELLENIC PARLIAMENT



ΙΔΡΥΜΑ
ΑΝΤΩΝΙΟΣ & ΙΩΑΝΝΗΣ
ΑΓΓΕΛΙΚΟΥΣΗΣ



ΔΗΜΟΣ Ierapetra.gov.gr
ΙΕΡΑΠΕΤΡΑΣ

ΙΕΡΑ ΜΗΤΡΟΠΟΛΙΣ
ΙΕΡΑΠΥΤΝΗΣ Κ ΣΙΤΕΙΑΣ

ΜΟΡΦΩΤΙΚΗ

ΣΤΕΓΗ ΙΕΡΑΠΕΤΡΑΣ

ΝΕΟΣ ΠΟΛΙΤΙΣΤΙΚΟΣ
ΣΥΛΛΟΓΟΣ ΙΕΡΑΠΕΤΡΑΣ



GOLDEN

ΠΛΑΣΤΙΚΑ
ΚΡΗΤΗΣ Α.Ε.

EXPENSES

24.950€

FTC Registration EuPE25, GrNC, EuPE26	3.875€
Equipment	6.000€
Airfaire Tickets	2.710€
Hotel	7.365€
Meals-Travel Transfers	5.000€

SILVER



BRONZE



SUPPORTERS



SUSTAINABLE